

AWS for GEN AI

Service	What It Does	Typical Use	Cost Type
Amazon Bedrock	Access managed foundation models (LLMs, image models) from providers like Anthropic, Meta, Amazon	Chatbots, summarization, RAG	Pay-per-token / image
Amazon S3	Object storage (e.g., datasets, model files, outputs)	Store PDFs, documents, model weights	Pay-per-GB stored + API calls
Amazon EC2	Rent virtual machines (CPUs or GPUs)	Run inference with open-source LLMs locally	Pay-per-hour (based on instance type)
Amazon SageMaker	Fully managed ML training + inference platform	Model fine-tuning, deployment, hosting	Pay-per-use (training, endpoints)
AWS Lambda	Serverless function runner (runs code on trigger)	Event-based tasks like summarizing on upload	Pay-per-millisecond of execution
Amazon API Gateway	Expose models as REST API endpoints	Create chatbot API with LLM as backend	Pay-per-call (free tier exists)
Amazon VPC	Isolate your infrastructure in the cloud	Secure LLM or RAG systems inside private networks	Free to use; pay for related services
AWS IAM	Controls who can do what	Secure access to Bedrock, EC2, etc.	Free
CloudWatch	Logs and monitors services	Debug or analyze model performance	Pay-per-metric, logs, and alarms
CloudFormation/CDK	Define infrastructure as code	Automate deployment of model pipelines	Free (you pay for provisioned resources)

Service	What It Does	Typical Use	Cost Type
AWS Glue	Serverless data integration and ETL	Preprocess datasets before model use	Pay-per-DPU-hour (for jobs)
Amazon EFS or EBS	Persistent storage attached to EC2	Store models or checkpoints	Pay-per-GB per month

Minimum for Interview + Hands-On:

Category	Service
Model Inference	Amazon Bedrock OR EC2
Storage	Amazon S3
Security	IAM
Infrastructure	EC2 + Launch Template + AMI
APIs	API Gateway + Lambda
Logs/Debug	CloudWatch
<i>(Optional but good)</i>	SageMaker (for training or hosting)

Cost-Saving Tips

- Use **SageMaker Studio Lab** (free Jupyter notebooks).
- **Spot Instances** for training (70% cheaper).
- **Lambda** for lightweight inference.